

Intelligent Transportation Systems (ITS) Analysis Modeling and Simulation (AMS) Community of Practice (CoP) – Spring 2026 Meeting

Wednesday April 22, 2026; 1:00 PM – 4:00 PM ET

Recording Information

Link	https://usdot.zoomgov.com/rec/share/cunkBgIzC7wJH-evEXIz1nkoIAuOE3lBhEt1m6PSj3VyxKAWGQwZRv0oN7CzQrt3.Q6vykiJ5erIACaxx
Passcode	zZH9%b6y

Attendees (55):

USDOT (10):	Mohamad Banihashemi (FHWA-HRSO), Ian Berg (Volpe), Andrew Berthaume (Volpe), Clayton Chen (FHWA-HRSO), James Colyar (FHWA-HOP), Anthony DePrator (FHWA-OH Division Office), John Hourdos (FHWA-HRSO), Gene McHale (FHWA-HRSO), Jongsun Won (FHWA-HOTM), Wei Zhang (FHWA-HRSO)
State/Local Agency (13):	Sana Arshad (Nevada DOT), Katie Beck (Michigan DOT), Shaza Elshazli (Nevada DOT), Eric Harmer (Nevada DOT), Thomas Hill (Florida DOT), Zhitong Huang (Virginia DOT), Mouyid Islam (Michigan DOT), Sanhita Lahiri (Virginia DOT), Chris Melson (Oregon DOT), Roxane Mukai (Maryland DOT), Mohammed Raqib (Maryland DOT), Rebekah Straub (Ohio DOT), Zifeng “Lilian” Wu (Caltrans)
Academia (7):	Marvin Baumann (Karlsruhe Institute of Technology), Angshuman Guin (Georgia Tech), Mohammed Hadi (Florida International), Samer Hamdar (George Washington University), Asad Khattak (Univ. of Tennessee), George List (NC State), Qing Tang (Old Dominion University)
Consultant (13):	Jessica Belowich (Michael Baker), Joseph Blasi (HNTB), Dante Bravo (Atkins), Jose Calvo (Leidos), Jordi Casas (Modexium), Ahmed Ghaly (Michael Baker), Amir Ghiasi (Leidos), David Hale (Leidos), Keir Opie (Transpo Group), Raj Paradkar (Kimley-Horn), Stanislav Parfonov (Michael Baker), Ossama Ramadan (WSP), Jaimison Sloboden (Michael Baker)
Vendor (2):	Chetan Joshi (PTV), Paulo Rinelli (Aimsun)
Non-USDOT Federal Partner (2):	Erin Boyd (DOE), Alexis Zubrow (DOE)
Noblis Support (6):	Asean Davis, Rick Grahn, Claire Silverstein, Kathy Thompson, Peiwei Wang, Karl Wunderlich
Unknown (2):	“PL953TH”, “GH”

Welcome and Recap

John Hourdos (FHWA)

Starts at 3:46 in recording

[John states purpose of community, provides recap from previous meeting, presents overview of the GitHub forum, and asks CoP members to provide email to join GitHub forum]

Community Member Announcements

CoP Members

Starts at 12:07 in recording

- **Jongsun Won (FHWA)** – Provided an update on the new practitioner Traffic Analysis Pooled Fund Study led by Colorado DOT that is being stood up. Noted that state and local DOTs are being encouraged to join and that there is still time for additional participants. [Pooled Fund Solicitation](#).
- **Samer Hamdar (GWU)** – Leading a project that is developing a tool with FHWA to test different CAV features and assess performance, with plans to share it with the broader community once complete. Also highlighted DDOT's work to establish a live ITS test bed in Southwest Washington, DC, where multiple companies will be invited to participate, with Samer/DDOT responsible for data collection, storage, and evaluation in alignment with this community.
- **Alexis Zubrow (DOE)** – Announced the DOE Annual Merit Review (June 1–4), which covers all DOE-funded projects and includes many relevant AMS topics this year. Noted that a database of past presentations is also available. [2026 Annual Merit Review](#).
- **Andrew Berthaume (Volpe)** – Developing a framework for CAV and traffic analysis and plans to recruit volunteers to gather input from. Interested individuals are encouraged to contact him about joining the working group.

Community Task Forces Organization, Mission, Activities

CoP Members

Starts at 19:52 in recording

Invited Presentations:

- **Chris Melson (Oregon DOT) [TDM/Long Range Planning]** – Highlighted a gap in national guidance and leadership in the TDM field and is working to address it through user-based groups and initiatives like the [Modeling Mobility Conference](#) and the Travel Demand Modeling Improvement Program. Noted that this Community of Practice (CoP) could help fill that gap, especially given the challenge of integrating ITS (typically 5–10-year horizons) into long-range planning models (20+ years). Emphasized difficulties in incorporating CAV behaviors and accounting for major economic and land use changes. Chris suggested involving FHWA's Office of Planning, Environment, and Realty (HEP) and engaging groups like the [Association of Metropolitan Planning Organizations](#). Also mentioned open-source tools like [ActivitySim](#) and [VisionEval](#), with a pooled fund underway to support the latter, as opportunities for this group to build upon.

- **John Hourdos (FHWA)** – Observed that automated vehicles may become widespread within 10–15 years, which is shorter than the lifespan of infrastructure being designed today (20+ year planning horizons). How should CAVs be considered?
- **Chris Melson (Oregon DOT)** – Agreed on the importance of integrating emerging technologies into planning and expressed interest in understanding potential large-scale behavioral shifts, such as changes in vehicle ownership, VMT, and parking demand, and how these should be reflected in TDM.
- **Jaimison Sloboden (Michael Baker)** – Stressed the need to improve current modeling practices, noting the difficulty of forecasting under varying growth scenarios and the importance of getting today’s models right first. There are too many uncertainties.
- **[From Chat] Alexis Zubrow (DOE)** – DOE’s Energy Efficient Mobility Systems (EEMS) program has done some modeling looking at the interaction of transportation and land use, specifically integrating POLARIS model with UrbanSim.
- **[From Chat] Jaimison Sloboden (Michael Baker)** – There is a whole spectrum from low/no growth areas to very high growth in future forecasts. Forecasts are often a reflection of policy, not just prediction.
- **[From Chat] Asad Khattak (U. of Tenn.)** – The Univ. of Tennessee-Knoxville and collaborating universities are working with rural planning organization and developed readiness tools and incorporating emerging technologies (latest is using a digital twin). Info about our work is here: [Tools for Assessing the Readiness of Rural Communities for New Technologies](#).
- **George List (NC State) [Network/Operations Models (Research)]** – The CoP can help develop consensus-based material that is also maybe software agnostic. Potential research opportunities include:
 - Calibration: Purpose sensitive, capital investments, operating changes, maintenance with different perspectives (top down, bottom up, new facility types).
 - Performance Metrics: Development, measurement, observation horizon and locations.
 - Demand Estimation: distinguishing between throughput and demand (best methods, in the field, simulation).
 - ODME: Methods (esp. with limited data).
 - Data Fusion: Non-concurrent data, expectations matching field performance.
 - Safety: How to represent/measure. Develop safety-related performance assessments, learning from vehicle trajectories/dynamics.
 - Geometric impacts: Impacts of geometry on behavior, performance.
 - Testing model performance: Robustness; methods to ensure this considering multiple operating conditions.
 - Examples of tool capability desires:
 - Geometry effects on behavior/performance, impacts on pricing, demand responsive to supply (latent demand), multimodal trips (transfers), closed-loop control, lane changing statistics, variance related metrics, among others.

- **Jaimison Sloboden (Michael Baker)** – Agreed the discussion is on target and plans to share additional needs offline. Mentioned an example of three freeway types (open, bridge, tunnel), noting that while models simplify these as links, such real-world conditions should be better represented in AMS because driver behaviors vary for these three freeway conditions.
- **John Hourdos (FHWA)** – Agreed with the points raised, particularly for simulation. ITS and connectivity require more research on how users respond to these technologies, since drivers ultimately decide whether to follow guidance (e.g., speed limits via connected vehicle messages). Highlighted differences between human and algorithmic behavior and not-yet-developed technologies, and the need for this group to lead in addressing these.
- **Anthony DePrator (FHWA)** – Raised the importance of incorporating ground-truth data, pointing out the lack of discussion around probe data and questioning how it can be leveraged at scale.
 - **George List (NC State)** – Explained that probe data helps reveal actual network performance and could be used to generate travel time distributions rather than relying solely on averages. Also noted that achieving reliable probe penetration depends on traffic flow, with lower volumes requiring longer observation periods to gather sufficient data.
 - **[From Chat] Mouyid Islam (Michigan DOT)** – Is there any “calibration” effort in the simulation models to replicate the “ground truth” or reality? Sometimes it could be difficult to adjust those or input those in the simulation environment. You touched on those. Thanks!
- **[From Chat] Asad Khattak (U. of Tenn.)** – George, very nice ideas! Increasingly, we are using SUMO, VENTOS, and CARLA.
- **Jongsun Won (FHWA) [Network/Operations Models (Practice)]** – Emphasized the need for clear, actionable steps for practitioners, including what scenarios to test and what data to use. Highlighted workforce challenges at state and local DOTs, such as who will carry out AMS work, how to train staff, and how to guide them toward sound methodologies. All of these topics can be addressed by the task force.
- **John Hourdos (FHWA) [AI/ML/Data Analytics]** - Noted that probe data currently represents only a small share of vehicles, but the growing presence of connected vehicle (CV) technology in new vehicles will significantly expand available data. The scale of this data will require AI/ML and automation for collection, processing, and application. Also pointed out that AI-driven traffic control strategies introduce additional uncertainty, making system behavior harder to predict. The task force could address how (1) AI/ML can support AMS, (2) how to incorporate/manage AI-driven systems that affect traffic, and (3) handle large-scale data. AI could also streamline the development of inputs for microsimulation models.
 - **[From Chat] Jordi Casas (Modexium)** - The robustness of the simulation models requires a complete traffic data repository 24h/365 day to capture different seasons, weather,

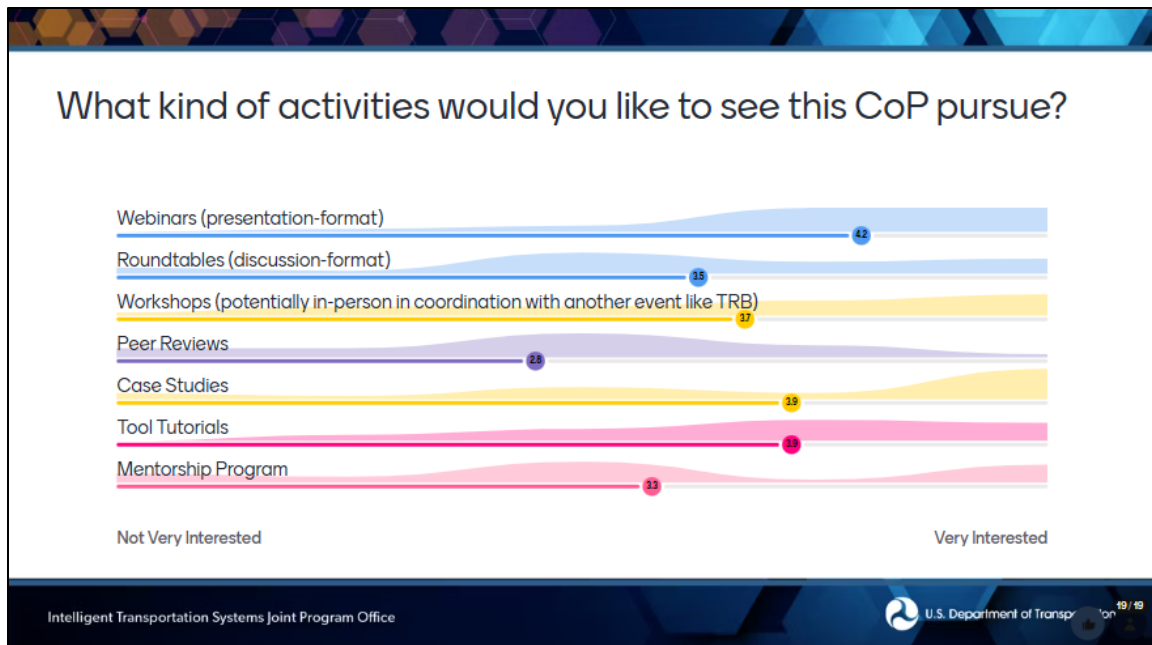
traffic regimes, etc. ... and ideally trajectory data. Are you aware about any initiative or existing resource with all this data available?

- **[From Chat] John Hourdos (FHWA)** - Methodologically the HCM travel time reliability analysis methodology has the right concepts to work this out, but it will need to be adapted to use the new data streams we have available. It is in my list of problems to investigate.
- **[From Chat] GH** – What about using connected vehicle data, e.g., from INRIX and Streetlight?
- **Ossama Ramadan (WSP) [Safety/Human Factors]** – There is a need to integrate safety into operations and simulation. This CoP/task force could help advance approaches that incorporate safety evaluation directly into AMS tools, which can help promote proactive use of simulation for future safety analysis (historically, safety professionals rely on lagging data, e.g., crash locations). There are also gaps in guidance, particularly for embedding safety performance into simulation inputs, and emphasized using trajectories to analyze near misses and conflicts. Recommended exploring AI-driven simulation and post-processing tools and stressed that safety evaluation should be integrated into all simulation studies, rather than treated as an afterthought. Concluded that the CoP can support practitioners by sharing knowledge, forming safety-focused groups, and coordinating with TRB to shape research needs.

Results from Mentimeter Q1 (30 participants):



Results from Mentimeter Q2 (19 participants):



Task Force Breakouts:

Starts at 1:31:18 in recording

Travel Demand Modeling/Scenario Planning

- **Breakout Members (6):** Andrew Berthaume, Chris Melson, Erin Boyd, Katie Beck, PL953TH, Rebekah Straub
- **Notetaker:** Karl Wunderlich
- **Draft Mission Statement:** Support the ITS AMS community in identifying needs related to the integration of CV, AV, and other ITS technologies into travel demand and scenario planning within ITS AMS.
 - Mission Statement is a decent catch-all, but will need some work in future, both title and mission statement.
 - Name: Are the two items [TDM and Scenario Planning] in the title related? First element is about the demand, the scenario part are the assumptions around the demand.
- Task Force Organization
 - Need a clear lead with bandwidth, and liaisons to other task forces and external groups
- Task Force Activities
 - Collect resources first, and let the content drive what the best engagement. Shout out for roundtables, where we can get many perspectives.
- Task Force Specific Meetings
 - Technology Transfer (T2) grants for peer exchanges, can request up to \$10k. We can find a number of state DOTs and FHWA. December is the target.
 - NCHRP synthesis activities – can these be leveraged? Note: These have not been funded (even when requested) in the past.

- Relationships with other groups
 - TMIP list serve still exists.
 - TRB is not looking for partnerships at the moment, and also not a one-to-one match with TRB after their reorganization.
 - National Operations Center of Excellence (NoCOE).

Network/Operations Models (Research)

- **Breakout Members (7):** Ahmed Ghaly, Angshuman Guin, George List, Jaimison Sloboden, James Colyar, Jessica Belowich, Marvin Baumann
- **Notetaker:** Claire Silverstein
- **Draft Mission Statement:** Support the ITS AMS community in identifying needs related to enhanced theory and algorithm development for models to effectively integrate, calibrate, and calculate impacts of CV, AV, and other ITS technologies using existing and new AMS approaches.
 - Looks okay, but there is room for improvement making the scope more achievable.
 - Consider adding forward thinking, gap filling, topics for research calibration, performance metrics, effectiveness of models (post review) scalability small to large.
 - Still don't have a mastery in simulation space to model human behavior, so bring back the following into the mix:
 - Mixed traffic conditions
 - Promote and improve human driven vehicles, pedestrians, and other modes and their interactions with CAVs.
 - Bring in clearinghouse framing and bring the human driver and human-driver component—need to interact with both.
 - The mission statement might need bullets to emphasize focus areas.
- Task Force Organization
 - Volunteer committees to focus on 1-2 initiatives at a time.
- Task Force Activities
 - Webinars are great – trying to raise the bar in terms of level of proficiency and professionalism that simulation projects are carried out. Roundtables are okay, workshops are possible, but could add those on to TRB committee activities. Peer reviews are a big deal to make sure things are acceptable to group. People who could present case studies would be great, same with tutorials. Mentorship program is interesting...could pair CoP folks who want to improve proficiency.
 - Target a group of states interested in sponsoring a pooled fund study – how to amass the resources to move the frontier? Who is the principal audience? States may be the audience that may work.
 - Propose NCHRP project?
 - AASHTO Standing Committee has other priorities and former TRB traffic simulation/modeling committee has been disbanded. FHWA might be willing to act as a proponent with simulation-friendly states to organize a pooled fund study to focus on topics of importance to these states. This could spill into an NCHRP proposal but need to have the support of simulation-friendly states to move the needle.
 - Model parameters will shift, but connectivity and levels of automation are not monolith, lots of variability.

- Does USDOT through NHTSA provide criteria that all have to follow? Or is there a range? Can a dataset be collected/derived from each manufacturer?
 - Need better understanding of state of practice and logic are for manufacturers – this is always evolving.
- What will be done that's not already being done? How to not duplicate efforts?
 - We could we serve as a framework/clearinghouse for research – then be agnostic about what's out there and just reflect as an expert panel to raise the flag for reporting out gaps/issues
 - Previous efforts made a matrix of models of what they can and can't do – should this task force do that for manufacturers on the CV/AV space with parameters and see what's acceptable/safe with different interactions?
- Theory development, develop case studies, field research, literature reviews.
- Task Force Specific Meetings
 - Coordinated with broader CoP meetings.
- Relationships with other groups
 - TRB, ITE, SIMCAP, AASHTO (they are more administrative/educational).

Network/Operations Models (Practice)

- **Breakout Members (7):** Ian Berg, Chetan Joshi, GH, Joseph Blasi, Jongsun Won, Keir Opie, Zifeng (Lilian) Wu
- **Notetaker:** Rick Grahn
- **Draft Mission Statement:** Support the ITS AMS community in determining the correct application and interpretation of models (micro, macro, meso) to effectively integrate, calibrate, and calculate impacts of CV, AV, and other ITS technologies using existing and new AMS approaches.
 - Replace “correct” with “appropriate”.
 - A focus should be on how to get information to the broader practitioner group considering existing tool limitations. Also, guidance related to inputs/parameters should be complemented with information about potential limitations/uncertainty of the assumptions.
- Task Force Organization [no feedback]
- Task Force Activities
 - Industry outreach – As a task force, reach out to OEMs/ADS developers to better understand technology capabilities (e.g., latency, feedback time)/behaviors, how to capture these capabilities in AMS, and communicate findings to practitioners.
 - Peer review would be valuable.
- Task Force Specific Meetings
 - Touchpoint maybe one month after the CoP meeting?
- Relationships with other groups
 - TRB – Committee on Intersection and Roadway Capacity and Quality of Service, Standing Technical Committee on Traffic Flow Theory, Models, and Simulation
 - ITE - Simulation and Capacity Analysis Committee (SIMCAP)

- [AASHTO - Committee on Transportation System Operations, Committee on Traffic Engineering](#)
- [Other - Pooled Fund Study – Traffic Analysis Roundtable](#)

AI/ML/Data Analytics

- **Breakout Members (8):** Amir Ghiasi, Anthony DePrator, Dante Perez-Brazo, David Hale, Mohammed Hadi, Zhitong Huang, Zihang Wei, John Hourdos
- **Notetaker:** Asean Davis
- **Draft Mission Statement:** Support the ITS AMS community in exploring how AI/ML can strengthen the phases of the modeling process for CV, AV and other ITS technologies, helping address current limitations and manage the growing complexity of modeling systems.
 - Expand the mission statement to include all AMS approaches including conventional simulation approaches and human driven vehicle considerations.
 - AI/ML can strengthen phases of the modeling, high-capacity analysis, etc. – this mission statement looks limited to CAV/AV technologies and should include all types of AMS.
- Task Force Organization
 - John to continue to lead and members can step in to assist.
 - Do we want the task force to just be a group of people who gather and discuss? Or should it be more organized with activities?
 - If we know what the end goal is it may be easier to set up the organization of the task force. Are we looking to do case studies, webinars, etc.?
 - Find items for community that are of interest and order them, then disseminate via webinars to get feedback.
- Task Force Activities
- Task Force Specific Meetings
 - Open to whatever setup is most productive. The listed activities are all valuable.
- Relationships with other groups
 - ITE, SITSO, SIMCAP
 - AASHTO CAV Practice Group – peer agencies meet every month or two.

Safety/Human Factors

- **Breakout Members (3):** Jose Calvo, Mohamad Banihashemi, Ossama Ramadan
- **Notetaker:** Kathy Thompson
- **Draft Mission Statement:** Support the ITS AMS community by identifying methods to integrate safety analysis and human-factor consideration into the assessment of CV, AV, and other ITS technologies to enhance transportation modeling.
 - [From GitHub]: Our mission is to strengthen ITS AMS by explicitly incorporating safety analysis, including related human factors, into assessments of various intelligent transportation systems and technologies. We focus on understanding interactions among road users, physical systems, and cyber systems in the context of a safe transportation system. Our work supports the development of safer and reliable ITS technologies for all users.

- Human factors is not often grouped with simulation. Human factors is an attribute used in safety analysis but is not the primary focus of safety analysis. Should be “Safety AMS”.
 - Where are human factors in the highway safety manual?
 - May want to identify some other attributes other than human factors.
- Suggested edits:
 - Change end to “to enhance transportation system safety” and not just “transportation modeling”
 - clarify that this is surface transportation, not open to aviation, other modes.
 - Change to “...to integrate safety analysis, human factors, driving algorithms into the assessment of....
- Task Force Organization
 - The task force should have a chair or co-chairs.
- Task Force Activities
 - First initiative will be recruiting more members. Interested in potentially doing webinars or workshops.
- Task Force Specific Meetings
 - Monthly/quarterly meetings in preparation for an upcoming event.
- Relationships with other groups
 - Ossama has contacts through ITE Data Driven Safety Analysis Committee, TRB Transportation Safety Management and Performance Committee
 - Human Factors Ergonomics Society (HFES)
 - TRB Human Factors Committee

Presentation: CAV Technology Tracking Resource Feedback

Noblis

Starts at 2:22:49 in recording

The following questions were posed to participants to gather feedback. Answers below were taken directly from the Mural Board [[Link](#)].

1. What types of AMS tools do you/your projects use?

John Hourdos (FHWA) – Microsimulation tools for operational analysis and ITS. Mesoscopic Activity Based models for long term prediction.

Jaimison Sloboden (Michael Baker) – Traditional modeling tools are used. We have also used economic analysis tools like [IMPLAN](#) to assess the benefits to smart freight logistics.

Chris Melson (Oregon DOT) – Tradition tools such as SIDRA, Synchro, VISSIM. Also use travel demand models: MPO models (activity-based) and simplified small urban models. Also have a statewide integrated model (with economic and land use).

Erin Boyd (DOE) – Agent-based transportation system models (POLARIS, BEAM CORE), which sometimes run linked to land use models like UrbanSim. Also use microsimulation models (SUMO, Vissim, Aimsun), 3D virtual environments (CARLA, IPG Carmaker).

Zhitong Huang (Virginia DOT) – Common microscopic simulation tools. Potentially use CARLA and NS-3 in the future.

Angshuman Guin (Georgia Tech) – Vissim and SUMO for simulation (and APIs in Vissim such as EBS and COM). Especially to study the interaction/aggression between autonomous vehicles and human

drivers. Reinforcement learning for traffic signal control, especially with respect to emergency vehicle preemption and transit signal priority.

Ossama Ramadan (WSP) – Microscopic simulation and deterministic tools for operational analysis. HSM-based tools for safety analysis.

Ian Berg (Volpe) – Strategic planning and scenario tools.

Thomas Hill (Florida DOT) – We make available to all the public agencies both macro- and microsimulation software packages. Training and understanding of model development and review processes are a statewide issue.

2. How do you consider or plan to consider CAVs in your future transportation plans?

Chris Melson (Oregon DOT) – Have basic guidance on when to consider CAVs in analysis. It is part of the Oregon DOT's Analysis Procedures Manual. It mainly pertains to when to use CAV penetration rate and how the penetration rate influences capacity.

Thomas Hill (Florida DOT) – There is an awareness that CAVs will have an impact but there is not much in the way of data to support how much. We developed statewide guidance on how to implement CAVs in travel demand models so that scenario testing doesn't overestimate the impacts.

Katie Beck (Michigan DOT) – Michigan State Long Range Plan (MM2045) considered CAV adoption in scenario planning ([link](#)). CAV adoption was a lever built into the state travel demand model.

Ossama Ramadan (WSP) – Typically through corridor-level ITS planning studies.

Zhitong Huang (Virginia DOT) – Yes, we are developing specific 5-year state-wide plans for CAVs.

John Hourdos (FHWA) – The actual driver behavior reaction to the information delivered is not yet known. How can we research this?

3. If you consider CAVs in your future plans, how do you currently represent emerging CAV technologies within the AMS tools for your projects?

Ossama Ramadan (WSP) – Vissim-based driver behavior models for CAVs.

Chris Melson (Oregon DOT) – Primarily related to how capacity is represented. The guidance we use is publicly available in Chapter 6 appendices of the Oregon DOT's Analysis Procedures Manual ([link](#)).

4. If you consider CAVs in your future plans, what are the most critical CAV data gaps for your specific AMS tasks?

Chris Melson (Oregon DOT) – Uncertainty on the assumptions to use beyond capacity adjustments.

Zhitong Huang (Virginia DOT) – AV trajectory data and how AVs interact with human driven vehicles and other road users.

Angshuman Guin (Georgia Tech) – AV trajectories and especially and response/interaction datasets with pedestrians, scooters, etc.

John Hourdos (FHWA) – Development of benchmarking datasets for the different driving behaviors expressed by the different drivers, human or not.

John Hourdos (FHWA) – How the different ADS control algorithms are implemented by the different manufacturers and how well they play together in the traffic stream.

5. Which is more useful: (a) vehicle make/model-level information or (b) aggregated fleet-level statistics?

Angshuman Guin (Georgia Tech) – (a) for researchers.

Thomas Hill (Florida DOT) – (b) for practitioners